

Food Access: Global to Alabama

Official-data pull, factual analysis, dataset sufficiency, and next-crawl recommendation

SCOPE: Global → United States → project-defined Southeast → Alabama. Food allergies are outside the active analysis and are handled as a separately gated next crawl.

Prepared for	Sharon Brooks, Sandra Kopecky, and April Gillespie
Frozen pull	August 22, 2026 (Central Time)
Primary local source	USDA ERS 2025 SNAP-authorized Retailer Access Map
Project region	AL, AR, FL, GA, KY, LA, MS, NC, SC, TN

Prepared as a factual checkpoint. The report does not assign causes, make causal claims, or combine unlike access measures into one score.

Executive summary

The pull supports the current main question. The primary local dataset is sufficient for Alabama and the ten-state comparison, provided the distance method and measurement boundaries remain explicit.

Key facts

- Global: FAOSTAT estimates 26.8% of the world population experienced moderate or severe food insecurity in 2023–2025. The U.S. FAOSTAT estimate is 10.7%.
- United States: USDA estimates 13.7% of U.S. households were food insecure in 2024—18.34 million households. Very low food security affected 5.4%, or 7.20 million households.
- Project-defined Southeast: 2022–2024 state household estimates range from 11.8% in North Carolina to 19.4% in Arkansas. Alabama is 12.1% $\pm$ 2.23 percentage points and ranks ninth of ten from high to low.
- Alabama: 20.2% of low-income residents—656,013 people—are beyond the primary retailer-access threshold.
- Alabama: 151 tracts, or 10.5%, are flagged low-income/low-access. The provisional U.S. share is 7.5%.
- Sensitivity: Alabama's low-income burden is 20.2% using driving distance and 8.6% using straight-line distance.

Decisions from this pull

Decision area	Recommendation	Reason
Primary local dataset	Retain USDA 2025 SRAM	Complete for Alabama and the ten project states; supports tract, population, low-income, and subgroup measures.
Global and U.S. context	Retain as separate layers	FAOSTAT population indicators and USDA household estimates answer different questions from retailer proximity.
More datasets	Add only against a declared gap	Avoid a broad crawl that mixes vintages and measures without an analytical purpose.
Allergies	Pause integration; run feasibility crawl	No single official public source links prevalence, tract food insecurity, store inventory, price, and allergens.

COMPARABILITY: Global population indicators, U.S. household food insecurity, and tract retailer proximity have different universes, units, and meanings. They are reported side by side, not as one numeric scale.

1. Data selected and captured

All retained sources are official, publicly downloadable, and frozen with URLs, file sizes, retrieval timestamps, and SHA-256 hashes. Raw files are reproducibly downloaded; processed tables and metadata are stored with the analysis.

Source ID	Dataset	Geography	Vintage
FAO_FS	FAOSTAT Suite of Food Security Indicators	Global, regional, and national	Current bulk release retrieved 2026-08-22
FAO_COAHD	FAOSTAT Cost and Affordability of a Healthy Diet	Global, regional, and national	July 2026 SOFI release
USDA_SRAM_2025	2025 SNAP-authorized Retailer Access Map	2020 U.S. census tracts	Retailers as of June 2025; released 2026-07-27
USDA_FOOD_SECURITY_CSV	Food Security in the United States data files	United States and states	2024 annual; 2022-2024 state average
USDA_ERR_358	Household Food Security in the United States in 2024 (ERR-358)	United States and states	Published 2025; state estimates average 2022-2024
CENSUS_TRACTS_AL_2020	2020 Alabama cartographic census tract boundaries, 1:500,000	Alabama 2020 census tracts	2020

Selection rules

- Use the latest official release available on the pull date; retain observation dates separately from retrieval dates.
- Keep a measure only when its unit, denominator, geography, and join key are clear.
- Use the 1-mile urban / 10-mile rural driving-distance threshold as the primary local measure because it matches the agreed plan and follows the road network.
- Preserve uncertainty for state household estimates and preserve alternate distance methods in sensitivity output.
- Do not average tract percentages. Aggregate numerators and denominators, then calculate the rate.

Capture package

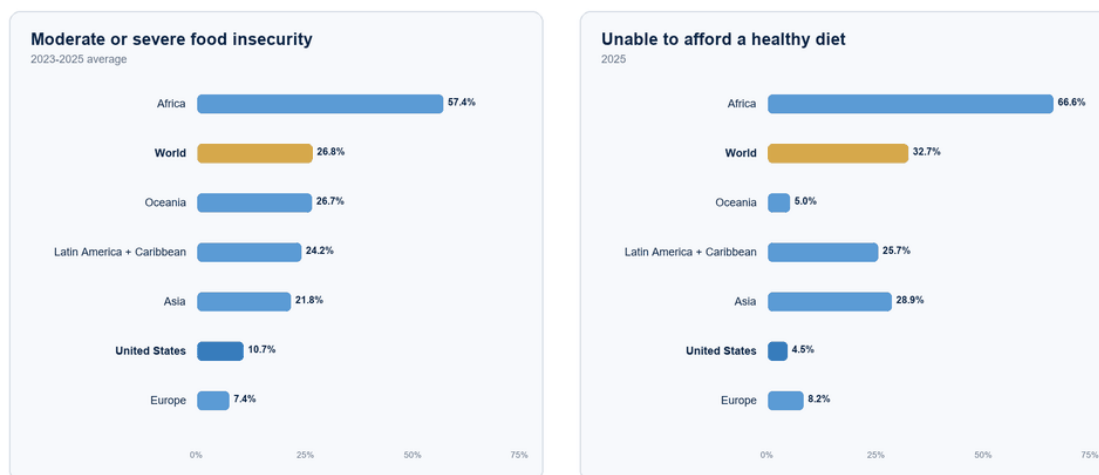
Artifact	Contents
Source manifest	Official URL, geography, vintage, retrieval time, bytes, SHA-256, local path.
Processed CSVs	Global summary, all state outcomes, all state retailer access, ten-state comparison, Alabama counties, all Alabama tracts.
Results	QA checks, sensitivity analysis, dataset sufficiency assessment, key findings.
Figures	Five reviewed charts/maps with source notes and consistent styling.
Team files	This report in DOCX/PDF and a filterable workbook with 12 sheets.
Reproducibility	Acquisition and analysis scripts plus dependency list and methods documentation.

2. Global access to food

FAOSTAT provides population-level indicators for food insecurity, undernourishment, and affordability of a healthy diet. Values below are the latest selected period for each series.

Global food-access indicators

Separate population measures; percentages should not be interpreted as the same construct.



Source: FAOSTAT Suite of Food Security Indicators and Cost and Affordability of a Healthy Diet, retrieved August 22, 2026.

Figure 1. Global food-access indicators from FAOSTAT. Series have different concepts but the same population-percent unit.

Geography	Moderate/severe FI	Severe FI	Undernourished	Unable to afford healthy diet
World	26.8%	9.8%	8.1%	32.7%
Africa	57.4%	21.1%	20.1%	66.6%
Asia	21.8%	8.6%	6.5%	28.9%
Europe	7.4%	1.9%	<2.5%	8.2%
Latin America and the Caribbean	24.2%	6.7%	5.1%	25.7%
Oceania	26.7%	10.3%	8.2%	5.0%
United States	10.7%	1.1%	<2.5%	4.5%

Periods: food-insecurity and undernourishment indicators use 2023–2025; healthy-diet unaffordability uses 2025. “<2.5” is preserved as published.

What stands out

- Africa has the highest selected moderate/severe food-insecurity estimate (57.4%), severe estimate (21.1%), undernourishment estimate (20.1%), and healthy-diet unaffordability estimate (66.6%).
- The world estimates are 26.8% moderate/severe food insecurity, 9.8% severe food insecurity, 8.1% undernourishment, and 32.7% unable to afford a healthy diet.
- The United States is below the world estimate on every selected FAOSTAT indicator: 10.7%, 1.1%, less than 2.5%, and 4.5%, respectively.

3. United States access to food

The U.S. section separates household food insecurity from physical proximity to SNAP-authorized retailers.

Household food security, 2024

Status	Share	Households
Food secure	86.3%	115.58 million households
Food insecure	13.7%	18.34 million households
Very low food security	5.4%	7.20 million households

Retailer access, primary threshold

Measure	Count	Share of matching denominator
Population beyond threshold	62,320,755	18.8%
Low-income population beyond threshold	49,667,746	21.2%
Low-income/low-access tracts	6,301	7.5%
Children beyond threshold	—	19.2%
Adults 65+ beyond threshold	—	20.4%
No-vehicle occupied units beyond threshold	—	8.9%
SNAP occupied units beyond threshold	—	11.1%

Primary threshold: more than 1 driving mile in urban tracts or 10 driving miles in rural tracts. The national low-income share is provisional because fourteen New York tracts lack low-income denominators.

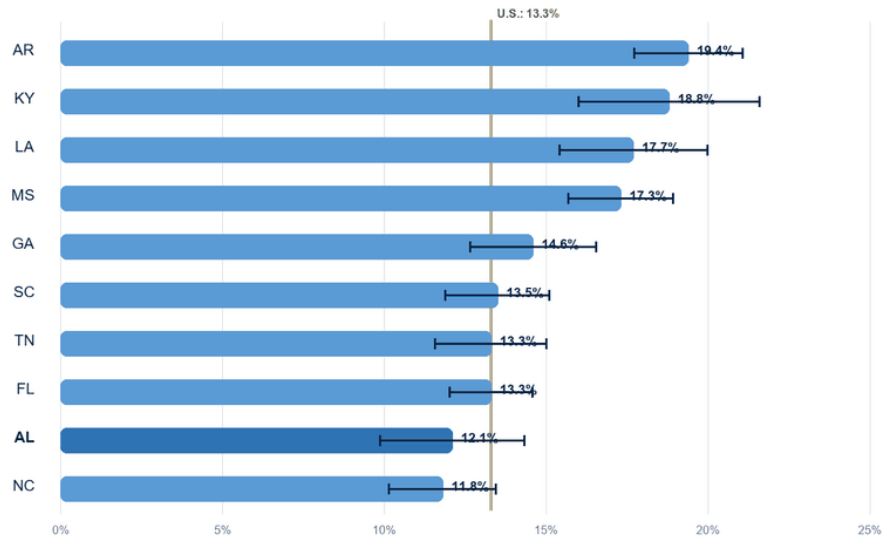
BOUNDARY: Household food insecurity is an outcome reported for households. SRAM retailer access is a tract-based proximity measure reported for people or housing units. A household can experience either, both, or neither.

4. Project-defined Southeast

The ten-state comparison uses the same state set throughout: Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, and Tennessee.

Household food insecurity in the project-defined Southeast

Average 2022-2024; error bars show USDA margins of error.



Source: USDA ERS, Food Security in the United States data files.

Figure 2. Household food insecurity in the project-defined Southeast, 2022–2024 average.

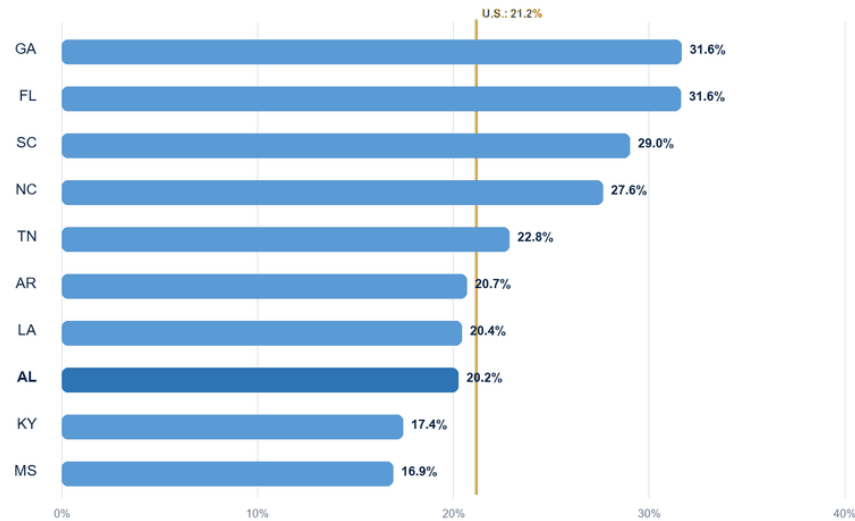
State	Food insecure	MOE (points)	Very low	Rank high→low
AR	19.4%	± 1.67	7.6%	1
KY	18.8%	± 2.80	9.0%	2
LA	17.7%	± 2.29	6.8%	3
MS	17.3%	± 1.62	6.9%	4
GA	14.6%	± 1.95	5.9%	5
SC	13.5%	± 1.60	5.8%	6
TN	13.3%	± 1.72	5.8%	7
FL	13.3%	± 1.28	4.8%	7
AL	12.1%	± 2.23	4.6%	9
NC	11.8%	± 1.66	4.4%	10

Margins of error are material. Small rank differences are descriptive, especially where intervals overlap.

5. Southeast retailer access

Low-income population beyond the primary retailer-access threshold

Driving distance: 1 mile urban / 10 miles rural; population-weighted state results.



Source: USDA ERS 2025 SNAP-authorized Retailer Access Map.

Figure 3. Low-income population beyond the primary retailer-access threshold.

State	Low-income burden	All-population burden	LILA tract share	Rank high→low
GA	31.6%	28.2%	13.6%	1
FL	31.6%	28.1%	10.1%	2
SC	29.0%	25.6%	10.8%	3
NC	27.6%	24.9%	12.0%	4
TN	22.8%	20.3%	9.8%	5
AR	20.7%	19.0%	16.8%	6
LA	20.4%	18.6%	11.8%	7
AL	20.2%	17.8%	10.5%	8
KY	17.4%	14.9%	6.8%	9
MS	16.9%	14.8%	10.4%	10

What stands out

- Georgia (31.6%) and Florida (31.6%) have the highest low-income retailer-access shares in the ten-state set; Mississippi is lowest at 16.9%.
- Alabama is 20.2%, eighth-highest of ten, and 0.9 percentage points below the provisional U.S. estimate of 21.2%.
- Household food-insecurity ordering and retailer-access ordering differ. Arkansas ranks first on the household outcome and sixth on retailer access; Georgia ranks fifth and first, respectively.

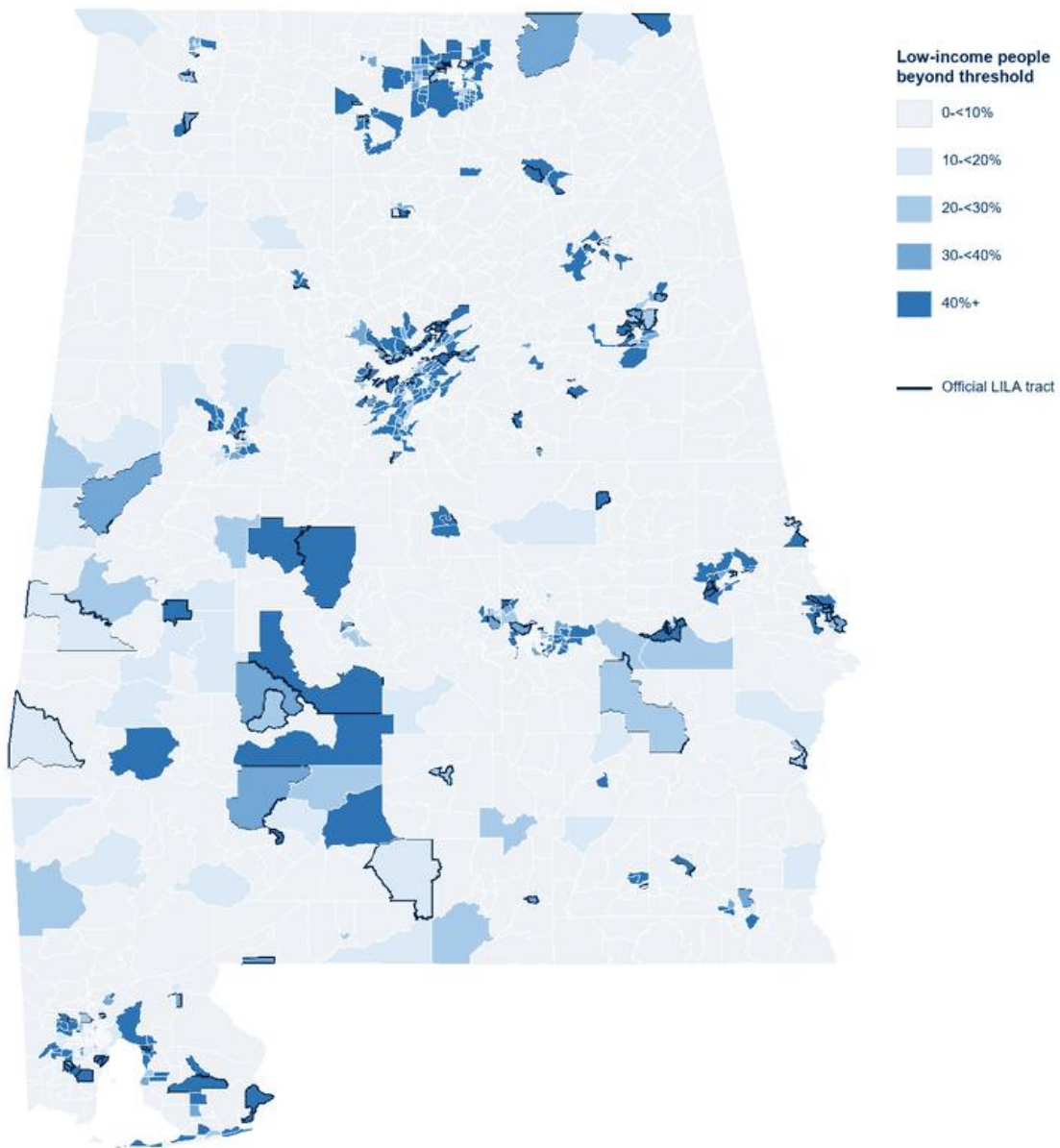
6. Alabama access to food

Alabama has complete primary fields for all 1,436 2020 census tracts and complete joins to the selected Census tract geometry.

Measure	Count	Share
2020 population	5,024,279	—
Low-income population	3,243,903	—
Population beyond primary threshold	895,423	17.8%
Low-income population beyond threshold	656,013	20.2%
Low-income/low-access tracts	151	10.5%

Alabama low-income population beyond the primary access threshold

Share of each tract's low-income population beyond 1 driving mile urban / 10 driving miles rural



Sources: USDA ERS 2025 SRAM; U.S. Census Bureau 2020 cartographic tract boundaries.

Figure 4. Alabama low-income retailer-access burden by census tract.

Dark shading indicates a larger share of the tract's low-income population beyond the threshold; the map does not show household food-insecurity prevalence.

7. Alabama county and tract detail

Counties below are ordered by the count of affected low-income residents, not by rate. This favors places where the absolute number is largest.

County	Affected low-income residents	Low-income burden	LILA tracts	LILA tract share
Jefferson	133,014	30.8%	33	17.5%
Madison	106,257	35.3%	8	8.4%
Shelby	68,184	36.7%	1	1.8%
Baldwin	42,834	23.9%	5	11.6%
Mobile	39,741	15.8%	13	9.6%
Lee	37,598	32.7%	10	20.8%
Tuscaloosa	36,811	25.1%	4	6.8%
Montgomery	33,781	25.4%	6	8.5%
Morgan	18,222	21.7%	2	6.5%
Calhoun	14,618	21.9%	10	26.3%

Highest affected counts

- Jefferson County: 133,014 affected low-income residents; 30.8% of its low-income population.
- Madison County: 106,257; 35.3%.
- Shelby County: 68,184; 36.7%.
- Baldwin County: 42,834; 23.9%.
- Mobile County: 39,741; 15.8%.

A high count and a high rate answer different questions. The workbook contains all 67 counties and both rankings.

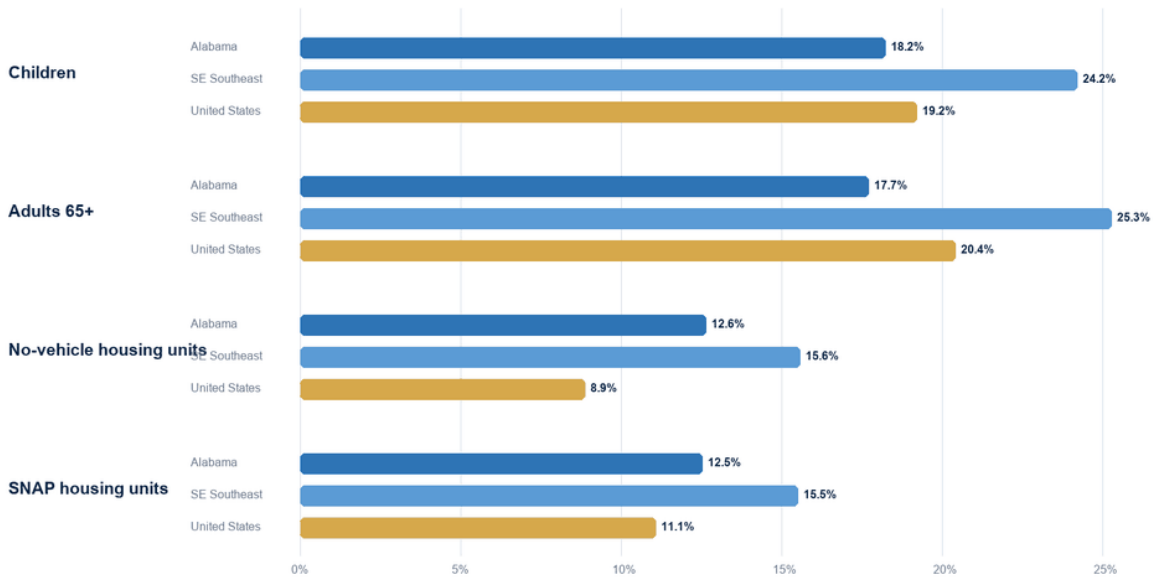
Top LILA tracts by affected low-income count

GEOID	County	Affected low-income	Low-income burden	Poverty rate
01073011110	Jefferson	3,388	78.6%	21.6%
01081040602	Lee	3,194	48.2%	23.5%
01003011000	Baldwin	3,003	80.1%	16.7%
01073012701	Jefferson	2,232	70.0%	14.7%
01073012905	Jefferson	2,136	50.7%	27.6%
01073011208	Jefferson	2,005	68.2%	19.3%
01089010626	Madison	1,717	65.0%	12.4%
01003011603	Baldwin	1,632	69.5%	3.5%

8. Alabama subgroup access

Primary-threshold subgroup access

Driving distance: 1 mile urban / 10 miles rural. Population and housing-unit denominators are kept separate.



Source: USDA ERS 2025 SNAP-authorized Retailer Access Map. Derived mixed urban/rural threshold counts.

Figure 5. Primary-threshold retailer access by population or housing-unit subgroup.

Group	Alabama	United States	Denominator
All population	17.8%	18.8%	People
Low-income population	20.2%	21.2%	Low-income people
Children	18.2%	19.2%	Children
Adults 65+	17.7%	20.4%	Adults 65+
No-vehicle occupied units	12.6%	8.9%	Housing units
SNAP occupied units	12.5%	11.1%	Housing units

What stands out

- Alabama is below the provisional U.S. share for all people, low-income people, children, and seniors.
- Alabama is above the U.S. share for no-vehicle occupied units (12.6% versus 8.9%) and SNAP occupied units (12.5% versus 11.1%).
- The no-vehicle and SNAP figures use housing-unit denominators; they should not be numerically pooled with population shares.

9. Sensitivity and sanity checks

Alabama sensitivity

Method	Threshold	All population	Low-income population	LILA tracts
Driving	0.5 urban / 10 rural	34.0%	35.9%	22.0%
Driving	1 urban / 10 rural	17.8%	20.2%	10.5%
Driving	1 urban / 20 rural	17.1%	19.6%	9.7%
Straight line	1 urban / 10 rural	7.3%	8.6%	3.6%

- Changing the urban threshold from 1.0 mile to 0.5 mile raises Alabama's low-income burden from 20.2% to 35.9%.
- Changing the rural threshold from 10 to 20 miles lowers the low-income burden only slightly, from 20.2% to 19.6%.
- Replacing driving distance with straight-line distance lowers the result from 20.2% to 8.6%.
- Recommendation: retain the network 1/10 threshold and label the method on every table, chart, and spoken finding.

QA result

QA STATUS: 12 PASS • 1 REVIEW • 0 FAIL

Status	Check	Observed	Note
PASS	SRAM normalized row count	84119	—
PASS	SRAM geography coverage	51	—
PASS	SRAM duplicate GEOID	0	—
PASS	GEOID normalization impact	15636	General characteristics dropped leading zeros for state FIPS codes below 10; zfill(11) restored matches.
PASS	Latest state food-security rows	52	—
PASS	DD_SRAM_LAPOP1_10 does not exceed POP2020	0	—
PASS	DD_SRAM_LALOW1_10 does not exceed TractLOWI	0	—
PASS	la_kids_mixed does not exceed TractKids	0	—
PASS	la_seniors_mixed does not exceed TractSeniors	0	—
PASS	la_hunv_mixed does not exceed TractHUNV	0	—
PASS	la_snap_mixed does not exceed TractSNAP	0	—
REVIEW	Primary measure completeness	15	Fourteen Suffolk County, New York tracts lack low-income denominators; one zero-population Massachusetts tract lacks the LILA flag. Alabama and the project-defined Southeast are complete.
PASS	Alabama geometry join coverage	1436	—

The one reviewed exception consists of fourteen Suffolk County, New York tracts without low-income denominators and one zero-population Massachusetts tract without a LILA flag. Alabama and the ten project states are complete.

10. What is available and whether to change datasets

Source / gap	Grain	Recommendation	Main limitation
FAO_FS	Global/region/country	KEEP - context only	Not comparable to tract retailer access; no U.S. state detail.
FAO_COAHD	Global/region/country	KEEP - context only	National and regional estimates; no U.S. state or tract detail.
USDA_FOOD_SECURITY_CSV	National/state	KEEP - outcome benchmark	State margins of error are material; no official tract outcome.
USDA_SRAM_2025	2020 census tract	KEEP - primary local dataset	Measures proximity to SNAP-authorized retailers, not price, quality, inventory, transit, or household food insecurity.
CENSUS_TRACTS_AL_2020	2020 census tract	KEEP - mapping support	Geometry only; must join on an 11-digit text GEOID.
USDA_LRAM_2019	2010 census tract	OPTIONAL NEXT CRAWL	Different retailer universe and tract base; use as a sensitivity comparison, not a time series.
USDA_FOOD_ENVIRONM ENT_ATLAS	County/state	ADD ONLY FOR A DECLARED GAP	Do not add broadly; retain only variables with a specific analytical purpose.
SUBSTATE_FOOD_INSECURITY_OUTCOME	County or smaller	GAP - evaluate modeled source	Needed only if the team wants to explain local food insecurity rather than local retailer access.

Recommendation

RETAIN THE CURRENT CORE: USDA SRAM is sufficient for the primary Alabama and Southeast physical-access question. FAOSTAT and USDA household estimates are sufficient as separate factual context and outcome benchmarks.

- Do not replace SRAM with the 2019 Large Retailer Access Map. The retailer universe and tract base differ; use LRAM only as a labeled sensitivity comparison.
- Do not add the full Food Environment Atlas by default. Add a small, predeclared variable set only if the team decides to examine price, assistance, store environment, or another explicit county-level gap.
- If the team wants a local food-insecurity outcome, evaluate a modeled county or tract source separately and document that it is not directly comparable to the official state household series.
- Preserve the current frozen pull. Refresh only with a new dated snapshot and a change log.

Known blind spots

- SRAM does not measure food price, product quality, nutrition, inventory, store hours, transit, disability access, online purchasing, or household food insecurity.
- The listed retailer universe is SNAP-authorized retailers as of June 2025 and excludes farmers markets and delivery routes.
- Observation periods differ across layers. This is a cross-source snapshot, not a synchronized time series.

11. Recommended next crawl for food allergies

Keep allergies outside the active analysis until feasibility is demonstrated. The public official sources identified here describe different parts of the problem and do not form a direct tract-level access dataset.

Module	Starting source	What it can support	Boundary
Taxonomy	FDA nine major allergens	Milk, egg, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, sesame.	Scope control; not prevalence or availability.
Diagnosed prevalence	CDC/NCHS 2024 NHIS	National adult/child estimates and selected demographic/urbanization comparisons.	Public-use geography does not support a reliable tract-level Alabama join.
Product labels	USDA FoodData Central branded foods	Ingredients, brand owner, category, GTIN/UPC; pilot text classification.	Not store inventory or price; allergen parsing requires validation.
Recall events	openFDA food enforcement	Reason, report date, status, distribution pattern, state when available.	Events are not prevalence or product availability.
Local availability and price	Retailer inventory source	Product × store × date × price, with stable product identifiers.	Not present in the official public sources reviewed; licensing/API gate.

Proceed / pause rule

PROCEED ONLY IF: A product-by-store-by-date source with price, defensible geographic coverage, stable product identifiers, and a validated FDA-allergen mapping is available. Otherwise, keep prevalence, product-label coverage, and recall events as separate descriptive modules or future work.

Recommended crawl sequence

- Lock the FDA nine-major-allergen taxonomy. Keep fruit and chocolate only as separately defined exploratory categories.
- Profile 2024 NHIS prevalence fields, denominators, demographic detail, and public-use geography.
- Pilot FoodData Central ingredient/Contains-text classification on a small sample and manually validate false positives and false negatives.
- Pull openFDA enforcement records for undeclared-allergen events; deduplicate and keep recall geography and dates.
- Search for a retailer inventory-and-price source only after the first four modules are documented; stop if product/store/date joins cannot be proven.

12. Reusable prompt for the allergy crawl

Conduct an official-source-only feasibility crawl for food-allergy access in Alabama and the project-defined Southeast (AL, AR, FL, GA, KY, LA, MS, NC, SC, TN). Keep this separate from the current food-access analysis until joinability is proven. Use the FDA nine major allergens—milk, egg, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, and sesame—as the primary taxonomy. Treat fruit and chocolate only as explicitly defined exploratory categories. For each candidate dataset, record publisher, exact dataset and table, URL/API endpoint, release and observation dates, population or product universe, geography, unit of analysis, allergen fields, price/inventory fields, identifiers, access/licensing limits, update frequency, missingness, and proposed joins. Prioritize CDC/NCHS 2024 NHIS for diagnosed prevalence, USDA FoodData Central branded foods for a pilot of ingredient/Contains-text classification, and openFDA food-enforcement data for undeclared-allergen recall events. Do not treat recalls as prevalence or FoodData Central as store inventory. Do not make causal or medical claims and do not collect personal health information. Sanity-check denominators, duplicate records, changing labels, geography, dates, and taxonomy. Deliver (1) a source inventory, (2) a field-level data dictionary, (3) small reproducible pilot extracts, (4) QA findings, (5) a joinability matrix, and (6) a proceed/pause recommendation. Proceed to local allergen-safe access only if a product-by-store-by-date source with price and defensible coverage is available; otherwise recommend separate descriptive modules and identify the exact missing data.

Expected output contract

Output	Acceptance check
Source inventory	Every row has an official owner, direct URL/API endpoint, vintage, geography, grain, and access terms.
Field dictionary	Denominators, identifier formats, allergen text fields, and missing-value conventions are explicit.
Pilot extracts	Small, reproducible, dated, hashed, and free of personal health information.
QA report	Duplicates, denominators, taxonomy, date ranges, geography, and label parsing are tested.
Joinability matrix	Every proposed join lists cardinality, expected match rate, temporal alignment, and failure conditions.
Decision	Proceed/pause is tied to product-by-store-by-date inventory and price coverage, not project novelty.

Appendix A. Processing and validation details

Tract key repair

The SRAM general-characteristics CSV stores CensusTract20 without leading zeros when imported as a number. Distance files preserve 11 digits. Normalizing every tract key to text with `zfill(11)` restored 15,636 joins, including all Alabama and Arkansas tracts. The pipeline then verified 84,119 unique U.S. tracts, 50 states plus the District of Columbia, and zero duplicate GEOIDs.

Aggregation

- Overall burden = sum of people beyond the selected threshold ÷ sum of 2020 population.
- Low-income burden = sum of low-income people beyond the selected threshold ÷ sum of low-income population.
- Subgroup shares use their matching child, senior, no-vehicle housing-unit, or SNAP housing-unit denominator.
- LILA tract share = flagged tracts ÷ all tracts in the geography.

Interpretation limits

- Descriptive comparisons do not establish causes or policy effects.
- State household margins of error should accompany estimates; ranks are descriptive.
- Physical proximity is one dimension of food access and cannot be read as product availability or affordability.
- Driving and straight-line results differ materially; method choice must be disclosed.
- The national low-income retailer-access estimate is provisional because of fourteen missing New York denominators. Alabama and the project region are unaffected.

Appendix B. Files delivered

Path	Purpose
reports/Data_for_Dinner_Food_Access_Analysis.docx	Editable team report.
reports/Data_for_Dinner_Food_Access_Analysis.pdf	Fixed-layout report.
reports/Data_for_Dinner_Food_Access_Analysis.xlsx	Filterable source, data, sensitivity, QA, and metadata workbook.
data/processed/*.csv	Analysis-ready tables at global, state, county, and tract levels.
data/metadata/*.csv	Source manifest and analysis variable dictionary.
results/*.csv / *.json	QA, sensitivity, sufficiency assessment, and key findings.
src/*.py	Reproducible acquisition and analysis.
docs/*.md	Methods and allergy next-crawl documentation.

Appendix C. Official sources

1. Food and Agriculture Organization of the United Nations. [FAOSTAT Suite of Food Security Indicators](#) Global, regional, and national food-security indicators.
2. Food and Agriculture Organization of the United Nations. [FAOSTAT Cost and Affordability of a Healthy Diet](#) Healthy-diet affordability indicators through 2025.
3. USDA Economic Research Service. [Household Food Security in the United States in 2024](#) 2024 national and 2022–2024 state household estimates.
4. USDA Economic Research Service. [Food Security in the U.S.: Key Statistics & Graphics](#) Official national summary and definitions.
5. USDA Economic Research Service. [Food Access Research Atlas — Download the Data](#) Official 2025 SRAM download and release information.
6. USDA Economic Research Service. [SRAM Reference Guide](#) Variables, thresholds, and retailer universe.
7. USDA Economic Research Service. [SRAM Data Sources and Technical Methods](#) STARS, Census, LandScan, ACS, distance, and method details.
8. U.S. Census Bureau. [2020 Alabama Census Tract Boundaries](#) Mapping geometry.
9. U.S. Food and Drug Administration. [Food Allergies](#) FDA major-allergen taxonomy and labeling context.
10. CDC National Center for Health Statistics. [2024 Adult Food-Allergy Estimates](#) Diagnosed adult prevalence.
11. CDC National Center for Health Statistics. [2024 Child Food-Allergy Estimates](#) Diagnosed child prevalence.
12. USDA Agricultural Research Service. [FoodData Central Data Dictionary](#) Branded-food product fields and ingredients.
13. U.S. Food and Drug Administration. [openFDA Food Enforcement API](#) Food recall enforcement events.

Appendix D. Final factual checkpoint

Check	Result
Main question answered without allergies	Yes
Global, U.S., Southeast, Alabama facts reported	Yes
Primary measure sufficient for project geography	Yes
Subgroup denominators clear	Yes
Driving-distance sensitivity disclosed	Yes
Source hashes and retrieval dates captured	Yes
Blocking QA failures	None
Allergy integration	Paused pending feasibility gate